Chapter 6: Research Plan Workshop

Preparing Your Research Plan

Gordon Wright

1 Overview

This reading prepares you for the lab session where you'll develop your Research Plan. This is a **workshop-based lab** — you'll be writing and drafting, not coding.

> **i What You'll Do**
>
> By the end of this lab, you'll have:
>
> - Operationalised your IV and DV for your chosen topic
> - Written H_0 and H_1 with appropriate direction justification
> - Decided between-subjects or within-subjects design
> - Matched your design to the appropriate statistical test
> - Drafted your Method section

2 Before the Lab

2.1 What to Bring

- Notebook or laptop for writing drafts
- Your lecture notes from Wednesday
- Access to the online lab activity (for MCQs)

2.2 What to Know

Make sure you're comfortable with these concepts:

1. **Operationalisation**: Turning abstract concepts into measurable variables
2. **Hypotheses**: H_0 (no effect) vs H_1 (effect exists); one-tailed vs two-tailed
3. **Designs**: Between-subjects (different people) vs within-subjects (same people)
4. **The test link**: Between → independent t-test; Within → paired t-test

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3 The Three Topics

Your Research Plan will address ONE of these questions:

3.1 Topic 1: Font Difficulty → Memory

Question: Does reading text in a harder-to-read font improve memory retention?

Background: Diemand-Yauman et al. (2011) proposed that "desirable difficulties" can enhance learning. Harder-to-read fonts require more cognitive effort, which might deepen processing and improve retention.

Key operationalisation decisions:

- What specific fonts will you use?
- How will you ensure only readability differs?
- How will you measure memory?

3.2 Topic 2: Hydration → Cognitive Performance

Question: Does drinking water improve cognitive task performance?

Background: Edmonds & Burford (2009) found that children who drank water performed better on cognitive tests. Dehydration may impair attention and processing speed.

Key operationalisation decisions:

- How much water? When?
- What's your control condition?
- What cognitive task will you use?

3.3 Topic 3: Eye Contact → Perceived Trustworthiness

Question: Does eye contact duration affect how trustworthy a person is perceived to be?

Background: Research on gaze and social perception shows that direct eye contact influences impressions of confidence, competence, and trustworthiness.

Key operationalisation decisions:

- Photos or videos?
- How will you manipulate gaze direction?
- How will you measure trustworthiness?

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4 Templates for the Workshop

4.1 IV Operationalisation Template

> "The independent variable is ___________ with two levels:
>
> **Condition 1**: [Specific details including exact parameters]
>
> **Condition 2**: [Specific details including exact parameters]"

Good example (caffeine study):

> "The independent variable is caffeine consumption with two levels:
>
> Condition 1 (Caffeine): 200mg caffeine capsule (equivalent to ~2 cups of coffee)
>
> Condition 2 (Placebo): Identical-looking capsule containing no caffeine"

Notice the specificity: exact dosage, matched appearance between conditions.

4.2 DV Operationalisation Template

> "The dependent variable is __________ measured by:
>
> **Task**: [What participants do]
>
> **Scoring**: [How performance is quantified]
>
> **Range**: [Possible values]"

Good example (caffeine study):

> "The dependent variable is alertness measured by:
>
> Task: Psychomotor Vigilance Test — respond to visual stimuli as quickly as possible
>
> Scoring: Mean reaction time across 30 trials (milliseconds)
>
> Range: Continuous, typically 200-500ms"

4.3 Hypothesis Template

Null hypothesis:

> H_0: There is no difference in [DV] between [Condition 1] and [Condition 2].

Alternative hypothesis (two-tailed):

> H_1: [DV] will differ between [Condition 1] and [Condition 2].

Alternative hypothesis (one-tailed):

> H_1: [DV] will be [higher/lower] in [Condition X] than [Condition Y].

4.4 Justification Template

> "I have chosen a [one-tailed / two-tailed] hypothesis because:
>
> Prior research [summary of relevant findings with citations].

Therefore, [I predict/do not predict] a specific direction."

4.5 Design & Test Justification Template

"This study employs a [between-subjects / within-subjects] design because [specific advantages for your study and how disadvantages are addressed].

Given this design, the data will consist of [two independent groups / paired scores from the same participants].

Therefore, an [independent samples / paired samples] t-test is appropriate because [explain how data structure matches test requirements]."

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5 Method Section Structure

Your Research Plan requires a brief Method section with four subsections:

5.1 Design

State the design type and define variables:

- Design type (between-subjects or within-subjects)
- IV with both levels fully operationalised
- DV with measurement details

5.2 Participants

Describe your sample:

- Target sample size
- Recruitment source
- How participants are assigned to conditions (or counterbalancing for within-subjects)

5.3 Materials

Describe everything participants will see or use:

- Stimuli (with specific details)
- Measures (rating scales, tests, etc.)
- Equipment (computers, software, paper materials)

5.4 Procedure

Describe what happens step by step:

- Setting (where testing occurs)
- Instructions given
- Sequence of events with timing
- Total session duration

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6 Checklist for Your Draft

Before leaving the lab, check that you have:

! Draft Checklist

Operationalisation:

- ☐ IV specified with exact parameters for both conditions
- ☐ DV specified with task, scoring, and range
- ☐ Another researcher could replicate from your description

Hypotheses:

- ☐ H_0 clearly stated (no effect)
- ☐ H_1 clearly stated (effect exists)
- ☐ Direction justified with reference to prior research

Design & Test:

- ☐ Design type stated and justified
- ☐ Disadvantages addressed
- ☐ Test selected and matched to design
- ☐ Explanation of WHY this test fits your data structure

Method Section:

- ☐ Design subsection drafted
- ☐ Participants subsection drafted
- ☐ Materials subsection drafted
- ☐ Procedure subsection drafted

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7 Common Mistakes to Avoid

7.1 In Operationalisation

- **Too vague**: "We showed participants different fonts" — which fonts exactly?
- **Confounded conditions**: Changing multiple things between conditions
- **Missing control details**: Only describing the experimental condition

7.2 In Hypotheses

- **No justification**: Choosing one-tailed without explaining why
- **Wrong direction**: Citing evidence for one direction but predicting the other
- **After-the-fact**: Planning to decide direction after seeing data (this invalidates one-tailed tests)

7.3 In Design Justification

- **Stating without explaining**: "I will use an independent t-test" with no justification
- **Mismatched design-test**: Claiming within-subjects but selecting independent t-test
- **Missing the logic**: Not explaining how design determines data structure

7.4 In Method Section

- **Not replicable**: Missing crucial details about timing, materials, or procedure
- **Too brief**: Participants, materials, and procedure each need substantive description
- **Missing counterbalancing**: For within-subjects designs, failing to address order effects

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8 Summary

Element	What to Include
IV Operationalisation	Variable name, both levels with exact parameters
DV Operationalisation	Variable name, task, scoring method, range
Null Hypothesis	Statement of no effect
Alternative Hypothesis	Statement of effect, with direction if appropriate
Direction Justification	Prior research supporting your choice
Design Justification	Advantages for your study, how disadvantages are managed
Test Justification	How your design produces data that matches the test

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9 Next Steps After the Lab

1. **Finish your draft** before Reading Week
2. **Seek feedback** from your tutor or study group
3. **Revise** based on feedback
4. **Submit** by Friday 27 February (Week 16)

⚠ Research Plan Due Date

Friday 27 February 2026, 12:00 noon

Allow time for revision. First drafts always need improvement.